

ABERDARE TECHNICAL COLLEGE
INFORMATION COMMUNICATION TECHNOLOGY
MODULE 2

UNIT: COMPUTER REPAIR AND MAINTENANCE |

UNIT CODE: 0714 351 04A

PRACTICAL 1, TERM 2 | LEVEL 5

Total:50 marks

Oral:30 marks

ASSESSOR'S OBSERVATION CHECKLIST

Trainer Details

Trainer name		Staff/ID no.	
Department		Date of assessment	
Trainer signature		Venue/Workstation	

Trainee Details

Trainee name		Admission no.	
Class/Course		Date	
Trainee signature		Workstation no.	

Instructions to the assessor:

- ✓ Observe the candidate throughout the practical and tick each criterion as it is met.
- ✓ Award marks up to the maximum shown.
- ✓ Part C (oral) is conducted after the practical.
- ✓ Enter marks in the summary table.

Cutting List: Tools and Materials

Items marked Centre are provided by the college. Items marked Student must be brought by the trainee.

No	Item	Quantity	Provided by
Hardware and consumables			

No	Item	Quantity	Provided by
1	Faulty desktop computer with keyboard, mouse, monitor and power cable	1 set	☐ Centre
2	Spare parts tray (1 HDD with OS installed, SATA data and power cables, other spare parts)	1 tray	☐ Centre
3	Diagnostic tools (bootable USB or DVD)	1	☐ Centre
4	Thermal paste	1 tube/sachet	☐ Centre
5	Cable ties and Velcro straps	10 ties, 2 straps	☐ Centre
Cleaning materials			
6	Air blower (hand blower or compressed air can)	1	☐ Centre
7	Microfibre cloth	1	☐ Centre
8	Soft anti-static brush	1	☐ Centre
9	Isopropyl alcohol (90% or higher)	1 small bottle	☐ Centre
10	Lint-free swabs/cotton buds	1 pack	☐ Centre
11	Dust mask	1	☐ Student
12	Safety goggles	1 pair	☐ Student
Tools			
13	Straight-head (flat) screwdrivers, large and small	1 each	☐ Centre
14	Phillips-head screwdrivers, large and small	1 each	☐ Centre
15	Torx screwdriver (if needed)	1	☐ Centre
16	Anti-static wrist strap	1	☐ Centre

No	Item	Quantity	Provided by
17	Anti-static mat	1	☐ Centre
18	Anti-static bags	2	☐ Centre
19	Needle-nosed pliers	1	☐ Centre
20	Tweezers or part retriever	1	☐ Centre
21	Wire cutters	1	☐ Centre
22	Chip extractor	1	☐ Centre
23	Plastic spudger/opening tool	1	☐ Centre
24	Magnetic parts tray or small labelled containers	1	☐ Centre
25	Flashlight or work lamp	1	☐ Centre
26	Marker pen and masking tape	1 each	☐ Centre
27	E-waste collection box	1 per room	☐ Centre
Documentation and stationery			
28	Computer/motherboard service manual	1	☐ Centre
29	Job card/repair log	1	☐ Centre
30	Pen	1	☐ Student
31	Ruler	1	☐ Student
32	Notepad	1	☐ Student

Part A: Practical Observation Checklist (50 marks)

No	Observation criteria (tick when observed)	Max	Marks	Remarks
Task 1: Preparation and Initial Inspection (6 marks)				
1	☐ Wears anti-static wrist strap clipped to bare metal of the case	2		
2	☐ Wears dust mask and safety goggles before blowing dust	1		
3	☐ Prepares a clean, well-lit workstation and unplugs the power cable before opening the case	1		
4	☐ Powers on, observes and records symptoms (fan noise, heat, BSOD, boot error, beep codes)	2		
Task 2: Disassembly and Diagnosis (12 marks)				
5	☐ Disconnects all external cables and peripherals	1		
6	☐ Removes side panel and photographs or sketches internal layout and cables	2		
7	☐ Identifies the faulty component using BIOS/UEFI, cable check and diagnostic tools, and records reasoning	4		
8	☐ Disconnects and removes the faulty part correctly, places it in an anti-static bag and labels it	3		
9	☐ Keeps screws organised and does not force parts	2		
Task 3: Cleaning and Dust Removal (13 marks)				
10	☐ Uses air blower correctly and holds fan blades still while blowing	3		

No	Observation criteria (tick when observed)	Max	Marks	Remarks
11	☐ Cleans CPU fan and heatsink fins thoroughly	2		
12	☐ Cleans case fans, vents, RAM and expansion slots	2		
13	☐ Cleans PSU vents from outside only and does not open the PSU	1		
14	☐ Checks thermal paste, cleans old paste and applies fresh pea-sized amount	3		
15	☐ Cleans monitor screen and casing with microfibre cloth, with no liquid sprayed on the screen	2		
Task 4: Installing the Replacement Part (8 marks)				
16	☐ Selects the correct replacement and checks compatibility (interface, size, connectors)	2		
17	☐ Mounts the replacement part in its position and secures it with screws	2		
18	☐ Connects data and power cables correctly and firmly	2		
19	☐ Ties cables neatly so no fan is blocked	2		
Task 5: Reassembly and Testing (8 marks)				
20	☐ Refits CPU cooler and fans, connects fan power, and closes the case	2		
21	☐ Reconnects peripherals, powers on and confirms POST with no error beeps	2		
22	☐ Enters BIOS/UEFI, confirms components are detected and sets the correct boot device	2		

No	Observation criteria (tick when observed)	Max	Marks	Remarks
23	☐ Verifies the repair (boots, no BSOD in about 10 minutes, fans and temperature normal)	2		
Task 6: Disposal and Final Clean-up (3 marks)				
24	☐ States correct disposal of the faulty part (data sanitization, certified e-waste recycler) and places it in the e-waste box	2		
25	☐ Returns all tools and leaves the workstation clean	1		
TOTAL PRACTICAL		50		

Part B: Product Checklist (10 marks)

No	Product criteria (tick when met)	Max	Marks	Remarks
1	☐ Computer boots to the desktop with no "No bootable device" message	3		
2	☐ No BSOD during the test run (about 10 minutes)	2		
3	☐ Fans spin freely, are dust-free, and temperature is in the normal range	2		
4	☐ Monitor screen and casing are clean and streak-free	1		
5	☐ Case closed neatly, no loose or missing screws and cables tidy	1		
6	☐ Job card fully and correctly completed (including the faulty component identified)	1		
TOTAL PRODUCT		10		

Part C: Oral Questions (30 marks)

Ask each question and award up to 2 marks (1 mark per correct key point).

Questions 1 to 8 come from Elements 2.1 and 2.2; Questions 9 to 15 are derived from the practical.

No	Question	Expected answer	Max	Marks	Remarks
2.1 Selection of computer components					

No	Question	Expected answer	Max	Marks	Remarks
1	<i>2.1.1 Computer hardware components</i> Name four internal hardware components of a desktop computer and state the function of each.	Any four with function (1 mark per two correct pairs): CPU (processes instructions); RAM (temporary working memory); motherboard (connects all components); HDD/SSD (permanent storage); PSU (supplies power); graphics card (video output); cooling fan/heatsink (removes heat).	2		
2	<i>2.1.1.1 Factors to consider</i> State four factors you consider when selecting a replacement component such as a hard disk.	Any four (half mark each): compatibility (interface e.g. SATA, form factor); capacity; speed/performance; power requirements; brand and quality; warranty; cost; availability.	2		
3	<i>2.1.1.2 Parts acquisition</i> State two ways of acquiring replacement computer parts and one precaution to take.	Ways: authorised dealers/retailers, online suppliers, manufacturer/vendor, salvaging from old machines, client supplying the part. Precautions: confirm genuine/new part, check warranty, keep receipt, test before fitting.	2		
2.2 Tools for repair and replacement					

No	Question	Expected answer	Max	Marks	Remarks
4	2.2.1.1 <i>Straight-head screwdriver</i> State one use of a large and a small straight-head (flat) screwdriver.	Large: loosening/tightening bigger slotted screws, e.g. case screws, prying covers. Small: small slotted screws in laptops/devices, releasing connector tabs and jumpers.	2		
5	2.2.1.2 <i>Phillips-head screwdriver</i> When would you use a large and a small Phillips-head screwdriver?	Large: case panels, PSU, drive-bay and motherboard standoff screws. Small: small drive, RAM and laptop screws, expansion cards. Matching size prevents stripped screw heads.	2		
6	2.2.1.3 <i>Tweezers/part retriever</i> What is a tweezers or part retriever used for in computer repair?	Picking up and placing small parts (screws, jumpers) and retrieving items dropped in tight spaces without touching the board.	2		
7	2.2.1.4 <i>Needle-nosed pliers</i> State two uses of needle-nosed pliers.	Any two: holding or bending small components, straightening bent pins, pulling out jumpers, reaching tight spaces, gripping small nuts/standoffs.	2		
8	2.2.1.5 <i>Wire cutters</i> and 2.2.1.6 <i>Chip extractor</i> State the use of wire cutters and of a chip extractor.	Wire cutters: cut cable ties and trim wires/cables. Chip extractor: safely removes IC chips from sockets without bending the pins (1 mark each).	2		
Questions derived from the practical					

No	Question	Expected answer	Max	Marks	Remarks
9	Why did you wear an anti-static wrist strap and unplug the computer before opening it?	Strap discharges static (ESD) that can damage chips; unplugging prevents electric shock and short-circuit damage.	2		
10	Explain how you identified the faulty component in this computer.	Linked symptoms (BSOD, "No bootable device") to storage; checked drive detection in BIOS/UEFI; checked cable connections; ran diagnostic/disk health tool; ruled out other parts.	2		
11	Why did you hold the fan blades still when using the air blower?	Blowing makes the fan over-spin and can damage the bearings and generate voltage that harms the board.	2		
12	Why did you apply thermal paste when refitting the CPU cooler?	Fills microscopic gaps for efficient heat transfer from CPU to heatsink; prevents overheating.	2		
13	How does dust accumulation cause overheating and how can it be prevented?	Dust blocks airflow and insulates heatsink so heat builds up. Prevention: regular cleaning, dust filters, clean room, proper case ventilation.	2		
14	What causes the BSOD and "No bootable device" error, and how did your repair fix it?	Failing or disconnected drive, corrupted system files or overheating. Replacing the drive with a working one containing a good operating system fixed the problem.	2		

No	Question	Expected answer	Max	Marks	Remarks
15	How did you dispose of the faulty part, and why is it not put in the ordinary bin?	Data wiped/destroyed, handed to certified e-waste recycler/collection box. Contains toxic metals (lead, mercury, cadmium) that pollute soil and water.	2		
TOTAL ORAL			30		

Summary of Marks

Section	Max marks	Marks awarded	Percentage
A. Practical observation	50		
B. Product	10		
C. Oral	30		
GRAND TOTAL	90		

Competence decision:

☐ Competent

☐ Not yet competent

Assessor's comments:

Candidate signature / Date	Assessor signature / Date